Methods description / MVA Hackathon 2026, Track 1 (Variant Prediction)

Please describe your model/approach in detail. A seven-stage Python pipeline using public REST APIs. (1) Gene shortlisting: eight HPO terms queried against the HPO JAX API, gene support weighted by ClinGen gene-disease validity, plus the three established MVA genes (BUB1B, CEP57, TRIP13). (2) Extract heterozygous PASS variants in those gene regions. (3) Per-variant Ensembl VEP annotation on MANE Select transcripts (SIFT/PolyPhen-2, ClinVar, gnomAD AF, domains). (4) NMD prediction for premature stops via the 5-prime exon-junction rule. (5) Optional AlphaGenome scoring (bonus only, never changes the ranking). (6) Filter (gnomAD AF < 1% or ClinVar P/LP) and rank with a fixed evidence table (LoF 80-100, damaging missense 50-70, other rare coding 10-40, x ClinGen weight), detecting compound-heterozygous pairs. (7) CSV output. Top finding: compound-heterozygous BUB1B c.2210T>G p.(Leu737Ter) (stop-gain, Pathogenic) + c.3006T>G p.(Asn1002Lys) (Likely Pathogenic if trans), epcr 0.85; trans inferred by a Bayesian posterior of 0.857-0.947.

(required) Generative AI used: provider, plan/tier, settings. Indiana University REALLMS (university self-hosted LLM endpoint) was used; non-commerical. This resource has been approved by the institution for use with PHI.

Automated output or manually reviewed/curated? Automated output of pipeline stage 7. The only manual input is the epcr parameter (0.85).

Describe any downstream manual review in detail. None on the variant content. The epcr (0.85) was set manually, conservatively below the 0.857-0.947 Bayesian trans posterior because phase is unmeasured.

Publicly available databases and literature only, or proprietary sources too? Public sources only + challenge-provided proband VCF.

Describe public data used in detail. GRCh38; Ensembl REST and VEP (MANE Select, ClinVar, gnomAD v4); HPO JAX API; ClinGen Knowledge Base validity grades; AlphaGenome (free registered API); AlphaMissense public scores (verification only, not in ranking); ACMG/AMP criteria from the published literature.

Describe any proprietary data used in detail. None.

Does your approach output compound-heterozygous pairs, or only single variants? Pairs: two damaging heterozygous variants in one gene are emitted on one submission row; our submission is such a pair.

How did you handle secondary or incidental findings? None; the submission contains a single primary row.

Estimated run time and cost. Under 30 minutes on a modern computer; zero cost (all free public APIs). Per-variant VEP queries dominate runtime.

Methods description / MVA Hackathon 2026, Track 2 (Drug Repurposing)

Describe your approach in detail (how you went from variant/mechanism to candidate medication(s)). A model-constrained repurposing screen. Starting from the Track 1 compound-heterozygous BUB1B pair (biallelic BUBR1 loss weakens the spindle-assembly checkpoint), a three-equation mass-action ODE model of the checkpoint (calibrated on BUBR1 perturbation data, independently validated, falsification documented) sizes the required effect (>= 2.2x functional BUBR1, conservative) and ranks levers (L1 BUBR1 amount = readthrough + hypomorph stabilization; L3 MCC stability, not druggable; L4 APC/C feedback, infeasible). Five filters: (1) lever - documented mechanism on L1; (2) effect size - published class effects must reach the model floor; (3) regulatory - approved in FDA/EMA/MHRA/PMDA with pediatric dosing experience, verified against primary sources (2026-10-15); (4) oncology - aneuploid-survival agents excluded (MVA1 is cancer-prone); (5) anti-target - checkpoint-impairing drugs (MPS1/TTK, Aurora B, PLK1, Eg5) excluded. Output: amlexanox (NMD inhibition + readthrough; the PTC sits 74 nt upstream of the final exon junction, so NMD is rate-limiting) plus Ravicti/4-PBA (pediatric chaperone for the destabilizing N1002K hypomorph) as a combination, because a dG bridge shows readthrough alone supplies at most ~28% of the required increment; ataluren is a readthrough-only comparator (regulatory footprint collapsing). A five-assay patient-cell validation plan with quantitative pass criteria is proposed (target: ~5-10x reduction of the aneuploidy rate in the dish).

(required) Generative AI used: provider, plan/tier, settings. Indiana University REALLMS (university self-hosted LLM endpoint) was used; non-commerical. This resource has been approved by the institution for use with PHI.

Was candidate identification automated or primarily manual? Hybrid: rule-based screening logic from the ODE model; automated ChEMBL target screen; manual final curation, and regulatory verification.

Describe any manual literature review or expert curation in detail. Primary-source regulatory audit (FDA/EMA/MHRA labels and ClinicalTrials.gov, verified 2026-10-15: ataluren EMA non-renewal and FDA NDA withdrawal; arimoclomol FDA approval and EMA refusal); literature searches for nonsense-suppression and chaperone effect sizes in patient cells from other diseases (CF, RDEB, DMD, ABCA1).

Publicly available databases and literature only, or proprietary sources too? Public sources only + challenge-provided proband VCF.

Describe the public data sources used in detail. ChEMBL (BUBR1 modulator screen - no approved or investigational scaffold/kinetochore modulator exists); ClinicalTrials.gov (amlexanox NCT01842282/NCT01975935); FDA/EMA/MHRA labels (DailyMed/EPAR/PLGB); OMIM and PubMed primary literature; AlphaMissense; ThermoMPNN stability dG; Boltz-2 co-complex structures; Pangolin regulatory-variant screen (heart/brain, no hits).

Describe any proprietary data sources used in detail. None.

How did you characterize the variant's mechanism? Biallelic BUB1B loss of function. Allele 1 c.2210T>G p.(Leu737Ter): stop-gain 74 nt upstream of the final exon junction, transcript destroyed by NMD - complete loss on that allele. Allele 2 c.3006T>G p.(Asn1002Lys): destabilizing hypomorph - BUB3/Cdc20 interface unchanged (Boltz-2) while stability is impaired (ThermoMPNN dG + 1.51 kcal/mol; AlphaMissense 0.9229) - partial loss (~15% residual function). Pathway disrupted: BUBR1/Bub3/MCC checkpoint inhibition of APC/C. Consequence: premature anaphase, chromosome mis-segregation, mosaic aneuploidy, cancer predisposition.

Estimated time or effort spent on this analysis. Several weeks of iterative work across the hackathon; cumulative ~70 hours.